MIL-S-45634A (MU)
3 January 1964
SUPERSEDING
MIL-S-45634 (ORD)
30 June 1959

MILITARY SPECIFICATION

SCALES, GRAPHICAL FIRING: M59, M60, M64, M70, M71, M74, M78, M79, M80, M81 AND M85

1. SCOPE

1.1 Scope. - This specification covers two groups of graphical firing scales identified herein as Type 1 and Type 2. The scales provide ballistic data and determine corrections pertinent to the fire of artillery weapons (see 6.1).

1.2 Classification.

1.2.1 Type 1 scales. - This type consists of graphical firing scales M59, M60, M64, M70, M71, and M85, each of which is composed of a rule and an indicator with hairline.

1.2.2 Type 2 scales. - This type consists of graphical firing scales M74, M78, M79, M80, and M81. Each scale is composed of a rule, an indicator assembly with hairline, and a slide.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-12937	Scales and Tables, Graphical Firing: Scales and Tables, Graphical Site; Packaging of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45208	Inspection Requirements, General Specification for

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U. S. Army Munitions Command

F7595189	Scale, Graphical Firing; M71
F7595188	Scale, Graphical Firing; M70
F7659932	Scale, Graphical Firing; M59
F7693988	Scale, Graphical Firing; M64
F8213754	Scale, Graphical Firing; M74
F8286783	Scale, Graphical Firing; M78
F8286784	Scale, Graphical Firing; M79
F8286785	Scale, Graphical Firing; M80
F8286786	Scale, Graphical Firing; M81
F8286787	Scale, Graphical Firing; M85
F8299947	Scale, Graphical Firing; M60

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. - Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication. - The scales shall be manufactured in accordance with the applicable drawings listed in Section 2 and all drawings pertaining thereto.

3.3 General specifications. - The contractor shall be responsible for adherence to and compliance with the following requirements of Specification MIL-F-13926.

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Part identification and marking.
(d) Workmanship.

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3.4 Environmental.

3.4.1 Operating temperatures. - The scales shall be operable when they are exposed to and thermally stabilized at ambient temperatures of -40° and +150°F.

3.4.2 Storage temperatures. - The scales shall meet the requirements of 3.5 when thermally stabilized at the standard ambient temperature of between 60° and 90°F after having been exposed to and thermally stabilized at ambient temperatures of -80° and +160°F.

3.4.3 Humidity. - The scales shall meet the humidity requirements of Specification MIL-F-13926, Procedure I. The scales shall then be thermally stabilized after return to the standard ambient temperature of between 60° and 90° F, and shall meet the requirements of 3.5.

3.5 Post-environmental. - Subsequent to the temperature and humidity tests specified in 3.4.2 and 3.4.3, the scales shall be operable and shall show no color changes, distortion, warpage, breaks, stickiness, cracks, mottling, or separation of bonded materials; and the scales shall meet all requirements of this specification.

3.6 Printing and surface quality. - Letters and numbers shall be in accordance with the applicable drawings. They shall be sharp, evenly colored, without skips or breaks, and shall not touch each other or any line of any scale. The printing surface shall be free from stains and other imperfections.

3.7 Scales. - All scales shall be printed in black or red as shown on pertinent drawings. The graduations shall be accurately placed so that error will not affect result of computations listed in 3.9 in excess of the tolerances indicated.

3.8 "C" and "D" scale graduation error - Type 2 scales.

3.8.1 With the first graduation of the "C" scale in coincidence with the first graduation of the "D" scale, each succeeding graduation on the "C" scale shall be in alinement with its respective graduation on the "D" scale within 0.005 inch.

3.8.2 A 2/3 ratio between graduations on the "C" and "D" scales shall be read throughout the scale within the tolerance indicated in 3.8.1 when the first graduation line on the "C" scale is set in coincidence with the 1.5 graduation on the "D" scale.

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3.9 Accuracy.

3.9.1 Type 1 scales. - Scale readings indicated in Tables I through IX shall be obtained, within the tolerances specified, for each respective range and condition listed.

3.9.2 Type 2 scales. - Deflection correction and quadrant elevation correction readings indicated in Tables X through XIV shall be obtained, within the tolerances specified, for the conditions listed.

TABLE I

SCALE, GRAPHICAL FIRING: M59

SHORT AND LONG RANGE FIRING DATA					
SCALE	RANGE (Meters)	ELEVATION (Mils)	FORK (F) (Mils)	DRIFT (Mils)	FUZE SETTING (FS) (Seconds)
Short Range	2,500	23.1 ± 0.1	0.9 ± 0.2	1.7 ± 0.2	4.6 ± 0.1
	4,500	58.1 ± 0.2	2.0 ± 0.2	3.9 ± 0.2	9.7 ± 0.1
Long Range	6,000	101.6 ± 0.5	3.5 ± 0.3	5.8 ± 0.3	14.9 ± 0.1
	8,000	186.6 ± 1.0	6.4 ± 0.4	8.9 ± 0.4	23.0 ± 0.1
	11,000	380.0 ± 2.0	14.8 ± 0.5	16.0 ± 0.5	38.7 ± 0.1
HIGH ANGLE FIRING DATA					
SCALE	RANGE (Meters)	ELEVATION (Mils)	10 m SI (Mils)	DRIFT (Mils)	TIME OF FLIGHT (Seconds)
High Angle	11,000	1092.2 ± 1.0	2.30 ± 0.05	64.6 ± 0.1	75.3 ± 0.1
	12,500	953.9 ± 1.0	6.1 ± 0.1	48.1 ± 0.1	68.8 ± 0.1

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TABLE II

SCALE, GRAPHICAL FIRING: M60

FIRING DATA				
W/O EXT ELEVATION	RANGE (Meters)	CHARGE RINGS	DRIFT (Mils)	TIME OF FLIGHT (Seconds)
800	1100 2000	5.68 ± 1/16 10.43 ± 1/16	48.8 ± 0.3 43.1 ± 0.3	15.3 ± 0.1 21.0 ± 0.1
900	1100 2000	5.826 ± 1/16 10.71 ± 1/16	55.9 ± 0.4 50.5 ± 0.4	16.9 ± 0.1 23.2 ± 0.1
1065	1100 2000	6.62 ± 1/16 12.2 ± 1/16	66.6 ± 0.5 59.3 ± 0.5	20.2 ± 0.1 27.8 ± 0.1

FIRING DATA				
W/EXT ELEVATION	RANGE (Meters)	CHARGE RINGS	DRIFT (Mils)	TIME OF FLIGHT (Seconds)
800	5200	39.0 ± 1/16	31.8 ± 0.3	35.4 ± 0.1
900	5200	40.39 ± 1/16	-----	39.2 ± 0.1

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TABLE III

SCALE, GRAPHICAL FIRING, M64 - RULE NO. 1

LOW ANGLE FIRING DATA			
SCALE	RANGE (Meters)	ELEVATION (Mils)	FUZE SETTING (Seconds)
Charge 1	1200 2500	146.0 ± 0.2 334.2 ± 0.5	6.17 ± 0.01 13.70 ± 0.02
Charge 3	2000 4000	148.5 ± 0.2 332.1 ± 0.5	8.10 ± 0.01 17.49 ± 0.03

HIGH ANGLE FIRING DATA					
SCALE	RANGE (Meters)	ELEVATION (Mils)	10 m/s SI (Mils)	DRIFT (Mils)	TF (Seconds)
Charge 1	3100	1129.2 ± 1.0	3.32 ± 0.02	46.08 ± 0.10	36.68 ± 0.02
Charge 2	4000	1104.4 ± 1.0	3.87 ± 0.03	43.42 ± 0.10	40.67 ± 0.02
Charge 3	5000	1110.8 ± 1.0	3.62 ± 0.03	44.08 ± 0.10	46.11 ± 0.02
Charge 4	6500	1095.8 ± 1.0	3.90 ± 0.03	42.55 ± 0.10	52.11 ± 0.03
Charge 5	8000	1113.8 ± 1.0	3.44 ± 0.03	44.41 ± 0.10	59.25 ± 0.03
Charge 6	10000	1114.5 ± 1.0	3.34 ± 0.03	44.48 ± 0.10	67.09 ± 0.03
Charge 7	12000	1134.0 ± 1.0	2.79 ± 0.02	46.65 ± 0.10	76.30 ± 0.04

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TABLE IV

SCALE, GRAPHICAL FIRING: M64 - RULE NO. 2

SCALE	RANGE (Meters)	ELEVATION (Mils)	FUZE SETTING (Seconds)
Charge 4	3500 5500	204.5 ± 0.2 360.3 ± 0.4	12.73 ± 0.01 21.69 ± 0.03
Charge 5	4000 7000	183.7 ± 0.2 376.0 ± 0.4	13.00 ± 0.01 25.28 ± 0.03
Charge 6	7000 9000	268.3 ± 0.2 391.0 ± 0.4	21.72 ± 0.02 30.17 ± 0.02
Charge 7	9000 12000	269.9 ± 0.3 444.2 ± 0.5	25.64 ± 0.02 38.62 ± 0.03

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TABLE V.

SCALE, GRAPHICAL FIRING M70 - RULE NO. 1

SCALE	RANGE (Meters)	QUADRANT ELEVATION (Mils) HEIGHT OF BURST SCALES									
		1000 Meters	950 Meters	900 Meters	850 Meters	800 Meters	750 Meters	700 Meters	650 Meters	600 Meters	Zero Height of Burst Scale (0)
Charge 3	1800	683.3 ± 0.5 ASC.	660.3 ± 0.5 ASC.	636.8 ± 0.4 ASC.	612.8 ± 0.4 ASC.	588.3 ± 0.4 ASC.	563.5 ± 0.4 ASC.	538.2 ± 0.3 ASC.	512.5 ± 0.3 ASC.	486.5 ± 0.3 ASC.	———
	4000	737.9 ± 0.7	717.4 ± 0.6	697.8 ± 0.6	678.5 ± 0.5	659.9 ± 0.5	641.8 ± 0.5	624.0 ± 0.5	606.5 ± 0.5	589.2 ± 0.5	395.4 ± 0.5
Charge 4	4000	563.4 ± 0.2	549.2 ± 0.2	535.3 ± 0.2	521.1 ± 0.2	507.2 ± 0.2	493.1 ± 0.2	479.1 ± 0.2	465.1 ± 0.2	450.9 ± 0.2	283.1 ± 0.3
	5500	732.3 ± 0.7	714.0 ± 0.6	696.6 ± 0.5	680.0 ± 0.5	664.2 ± 0.5	648.7 ± 0.5	633.9 ± 0.5	619.2 ± 0.5	605.1 ± 0.4	450.0 ± 0.4

SCALE	RANGE (Meters)	FUZE SETTING (FS) (Seconds) HEIGHT OF BURST SCALES								
		1000 Meters	950 Meters	900 Meters	850 Meters	800 Meters	750 Meters	700 Meters	650 Meters	600 Meters
Charge 3	1800	9.85 ±0.02	9.67 ±0.02	9.50 ± 0.02	9.34 ± 0.02	9.18 ± 0.02	9.04 ± 0.02	8.90 ± 0.02	8.77 ±0.02	8.65 ±0.02
	4000	24.28 ±0.04	23.84 ±0.04	23.44 ± 0.04	23.07 ±0.04	22.74 ±0.03	22.43 ±0.03	22.13 ±0.03	21.86 ±0.03	21.61 ±0.03
Charge 4	4000	18.53 ±0.02	18.37 ±0.02	18.22 ±0.02	18.07 ±0.02	17.92 ±0.02	17.80 ±0.02	17.67 ±0.02	17.54 ±0.02	17.42 ±0.02
	5500	30.17 ±0.04	29.68 ±0.03	29.24 ±0.03	28.85 ±0.03	28.50 ±0.03	28.15 ±0.03	27.84 ±0.03	27.55 ±0.03	27.28 ±0.03

NOTE: Definition - ASC = Ascending trajectory.

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TABLE VI

SCALE, GRAPHICAL FIRING, M70 - RULE NO. 2

SCALE	RANGE (Meters)	QUADRANT ELEVATION (Mils)									
		HEIGHT OF BURST SCALES									
		1000 Meters	950 Meters	900 Meters	850 Meters	800 Meters	750 Meters	700 Meters	650 Meters	600 Meters	Zero Height of Burst Scale (0)
Charge 5	4500	479.2 ± 0.1	467.6 ± 0.1	455.8 ± 0.2	443.8 ± 0.2	431.8 ± 0.2	419.8 ± 0.2	408.0 ± 0.2	396.1 ± 0.2	384.2 ± 0.2	240.5 ± 0.3
	7000	679.8 ± 0.6	667.0 ± 0.6	654.5 ± 0.6	642.4 ± 0.6	630.6 ± 0.6	619.0 ± 0.6	607.7 ± 0.6	596.6 ± 0.6	585.7 ± 0.6	464.5 ± 0.5
Charge 7	6500	363.9 ± 0.2	355.8 ± 0.2	347.7 ± 0.2	339.5 ± 0.2	331.5 ± 0.2	323.5 ± 0.2	315.4 ± 0.2	307.3 ± 0.2	299.1 ± 0.3	202.1 ± 0.3
	10500	600.7 ± 0.6	593.8 ± 0.6	586.8 ± 0.6	579.8 ± 0.6	573.1 ± 0.6	566.3 ± 0.6	559.5 ± 0.6	553.0 ± 0.6	546.4 ± 0.6	470.6 ± 0.5

SCALE	RANGE (Meters)	FUZE SETTING (FS) (Seconds)									
		HEIGHT OF BURST SCALES									
		1000 Meters	950 Meters	900 Meters	850 Meters	800 Meters	750 Meters	700 Meters	650 Meters	600 Meters	
Charge 5	4500	17.58 ± 0.02	17.48 ± 0.02	17.38 ± 0.02	17.29 ± 0.02	17.19 ± 0.02	17.11 ± 0.02	17.02 ± 0.02	16.94 ± 0.02	16.86 ± 0.02	
	7000	33.45 ± 0.04	33.10 ± 0.04	32.79 ± 0.04	32.50 ± 0.04	32.22 ± 0.04	31.96 ± 0.03	31.72 ± 0.03	31.49 ± 0.03	31.26 ± 0.03	
Charge 7	6500	19.66 ± 0.03	19.59 ± 0.03	19.54 ± 0.03	19.49 ± 0.03	19.45 ± 0.03	19.40 ± 0.03	19.36 ± 0.03	19.32 ± 0.02	19.28 ± 0.02	
	10500	40.60 ± 0.05	40.43 ± 0.05	40.27 ± 0.05	40.11 ± 0.05	39.96 ± 0.04	39.81 ± 0.04	39.67 ± 0.04	39.53 ± 0.04	39.41 ± 0.04	

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TABLE VII

SCALE, GRAPHICAL FIRING: M71 - RULE NO. 1

LOW ANGLE FIRING DATA			
SCALE	RANGE (Meters)	ELEVATION (Mils)	FUZE SETTING (Seconds)
Charge 1	2000 4000	169.6 ± 0.2 381.2 ± 0.5	8.53 ± 0.01 18.56 ± 0.03
Charge 3	3000 5000	174.6 ± 0.2 317.2 ± 0.4	10.68 ± 0.01 18.92 ± 0.02

HIGH ANGLE FIRING DATA					
SCALE	RANGE (Meters)	ELEVATION (Mils)	10 m SI (Mils)	DRIFT (Mils)	TF (Seconds)
Charge 1	4500	1118.9 ± 1.0	3.35 ± 0.02	41.41 ± 0.08	43.68 ± 0.02
Charge 2	5500	1092.8 ± 1.0	3.96 ± 0.02	39.69 ± 0.08	47.03 ± 0.02
Charge 3	6500	1104.6 ± 1.0	3.55 ± 0.02	41.79 ± 0.08	52.08 ± 0.02
Charge 4	8000	1094.9 ± 1.0	3.66 ± 0.02	42.29 ± 0.08	57.44 ± 0.03
Charge 5	9500	1112 .7 ± 1.0	3.11 ± 0.02	46.43 ± 0.10	64.40 ± 0.02
Charge 6	12000	1069.2 ± 1.0	4.02 ± 0.03	44.32 ± 0.10	69.67 ± 0.03
Charge 7	14500	1070.9 ± 1.0	3.66 ± 0.03	47.31 ± 0.10	78.13 ± 0.03

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TABLE VIII

SCALE, GRAPHICAL FIRING, M71 - RULE NO. 2

SCALE	RANGE (Meters)	ELEVATION (Mils)	FUZE SETTING (Seconds)
Charge 4	4000 7500	193.5 ± 0.2 426.7 ± 0.5	13.14 ± 0.01 27.63 ± 0.03
Charge 5	5000 9000	190.3 ± 0.2 422.0 ± 0.5	15.00 ± 0.01 30.66 ± 0.03
Charge 6	7500 10500	239.0 ± 0.2 400.9 ± 0.4	21.32 ± 0.02 33.06 ± 0.03
Charge 7	9000 13000	216.9 ± 0.2 409.2 ± 0.4	22.91 ± 0.02 38.41 ± 0.03

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TABLE IX

SCALE, GRAPHICAL FIRING, M85

CHARGES 1, 2, AND 3			
SCALE	RANGE (Meters)	ELEVATION (Mils)	FUZE SETTING (Seconds)
Charge 1	2500 4000	213.0 ± 0.2 380.3 ± 0.5	10.59 ± 0.01 18.45 ± 0.02
Charge 2	3500 6000	162.5 ± 0.2 318.2 ± 0.4	11.12 ± 0.01 20.97 ± 0.03
Charge 3	7000 11000	183.6 ± 0.2 382.1 ± 0.5	17.85 ± 0.02 33.22 ± 0.04

MINIMUM ELEVATION SCALE			
RANGE (Meters)	ELEVATION (Mils)		
	Charge 1	Charge 2	Charge 3
500	39.1 ± 0.1	19.90 ± 0.06	8.60 ± 0.03
1500	121.8 ± 0.2	63.10 ± 0.07	27.30 ± 0.03

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TABLE X

SCALE, GRAPHICAL FIRING: M74

DEFLECTION CORRECTION COMPUTATIONS			
SLIDE FACE	WIND DIRECTION AND VELOCITY (Set "C" scale index over wind velocity value on "D" scale)	MET CORRECTED QE (Mils) (Set hairline of indicator assembly over met corrected QE value on "Lateral Wind Elevation" scale)	DEFLECTION CORRECTION (Mils) (Read on "D" scale under hairline)
Nighttime	Right 2 mph	300	-11.74 ± 0.03
Nighttime	Left 8 mph	650	+69.87 ± 0.13
All other	Right 12 mph	250	-56.16 ± 0.10
All other	Left 8 mph	660	+52.97 ± 0.10

QE CORRECTION COMPUTATIONS			
SLIDE FACE	WIND DIRECTION AND VELOCITY (Set "C" scale index over wind velocity value on "D" scale)	MET CORRECTED QE (Mils) (Set hairline of indicator assembly over met corrected QE value on appropriate "wind elevation" scale)	QE CORRECTION (Mils) (Read on "D" scale under hairline)
Nighttime	Head 3 mph	300 (On "Head Wind Elevation" Scale)	+4.85 ± 0.01
Nighttime	Head 9 mph	650 (On "Head Wind Elevation" Scale)	+41.77 ± 0.08
Nighttime	Tail 4 mph	350 (On "Tail Wind Elevation" Scale)	-7.38 ± 0.02
Nighttime	Tail 8 mph	700 (On "Tail Wind Elevation" Scale)	-38.08 ± 0.07
All other	Head 6 mph	660 (On "Head Wind Elevation" Scale)	+21.28 ± 0.04
All other	Head 20 mph	300 (On "Head Wind Elevation" Scale)	+26.88 ± 0.05
All other	Tail 15 mph	400 (On "Tail Wind Elevation" Scale)	-26.70 ± 0.05
All other	Tail 18 mph	750 (On "Tail Wind Elevation" Scale)	-69.31 ± 0.13

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TABLE XI

SCALE, GRAPHICAL FIRING, M78

DEFLECTION CORRECTION COMPUTATIONS			
SLIDE FACE	WIND DIRECTION AND VELOCITY (Set "C" scale index over wind velocity value on "D" scale)	MET CORRECTED QE (Mils) (Set hairline of indicator assembly over met corrected QE value on "Lateral Wind Elevation" scale)	DEFLECTION CORRECTION (Mils) (Read on "D" scale under hairline)
Nighttime	Right 2 mph	300	-11.32 ± 0.03
Nighttime	Left 8 mph	650	+64.09 ± 0.12
All other	Right 12 mph	400	-60.53 ± 0.12
All other	Left 8 mph	660	+48.45 ± 0.09

QE CORRECTION COMPUTATIONS			
SLIDE FACE	WIND DIRECTION AND VELOCITY (Set "C" scale index over wind velocity value on "D" scale)	MET CORRECTED QE (Mils) (Set hairline of indicator assembly over met corrected QE value on appropriate "wind elevation" scale)	QE CORRECTION (Mils) (Read on "D" scale under hairline)
Nighttime	Head 3 mph	300 (On "Head Wind Elevation" Scale)	+4.35 ± 0.01
Nighttime	Head 9 mph	650 (On "Head Wind Elevation" Scale)	+38.80 ± 0.07
Nighttime	Tail 4 mph	350 (On "Tail Wind Elevation" Scale)	-6.71 ± 0.02
Nighttime	Tail 8 mph	700 (On "Tail Wind Elevation" Scale)	-35.50 ± 0.07
All other	Head 6 mph	660 (On "Head Wind Elevation" Scale)	+19.76 ± 0.04
All other	Head 20 mph	300 (On "Head Wind Elevation" Scale)	+24.13 ± 0.05
All other	Tail 15 mph	400 (On "Tail Wind Elevation" Scale)	-24.75 ± 0.05
All other	Tail 18 mph	750 (On "Tail Wind Elevation" Scale)	-64.84 ± 0.12

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TABLE XII

SCALE, GRAPHICAL FIRING: M79

DEFLECTION CORRECTION COMPUTATIONS			
SLIDE FACE	WIND DIRECTION AND VELOCITY (Set "C" scale index over wind velocity value on "D" scale)	MET CORRECTED QE (Mils) (Set hairline of indicator assembly over met corrected QE value on "Lateral Wind Elevation" scale)	DEFLECTION CORRECTION (Mils) (Read on "D" scale under hairline)
Nighttime	Right 2 mph	300	-11.31 ± 0.03
Nighttime	Left 8 mph	650	+68.20 ± 0.13
All other	Right 12 mph	250	-54.05 ± 0.10
All other	Left 8 mph	660	+51.67 ± 0.10

QE CORRECTION COMPUTATIONS			
SLIDE FACE	WIND DIRECTION AND VELOCITY (Set "C" scale index over wind velocity value on "D" scale)	MET CORRECTED QE (Mils) (Set hairline of indicator assembly over met corrected QE value on appropriate "wind elevation" scale)	QE CORRECTION (Mils) (Read on "D" scale under hairline)
Nighttime	Head 3 mph	300 (On "Head Wind Elevation" Scale)	+4.89 ± 0.01
Nighttime	Head 9 mph	650 (On "Head Wind Elevation" Scale)	+41.30 ± 0.08
Nighttime	Tail 4 mph	350 (On "Tail Wind Elevation" Scale)	-7.31 ± 0.02
Nighttime	Tail 8 mph	700 (On "Tail Wind Elevation" Scale)	-37.83 ± 0.07
All other	Head 6 mph	660 (On "Head Wind Elevation" Scale)	+21.03 ± 0.04
All other	Head 20 mph	300 (On "Head Wind Elevation" Scale)	+27.12 ± 0.05
All other	Tail 15 mph	400 (On "Tail Wind Elevation" Scale)	-26.46 ± 0.05
All other	Tail 18 mph	750 (On "Tail Wind Elevation" Scale)	-68.88 ± 0.13

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TABLE XIII

SCALE, GRAPHICAL FIRING; M80

DEFLECTION CORRECTION COMPUTATIONS			
SLIDE FACE	WIND DIRECTION AND VELOCITY (Set "C" scale index over wind velocity value on "D" scale)	MET CORRECTED QE (Mils) (Set hairline of indicator assembly over met corrected QE value on "Lateral Wind Elevation" scale)	DEFLECTION CORRECTION (Mils) (Read on "D" scale under hairline)
Nighttime	Right 2 mph	300	-10.38 ± 0.02
Nighttime	Left 8 mph	650	+58.88 ± 0.10
All other	Right 12 mph	250	-49.95 ± 0.09
All other	Left 8 mph	660	+44.52 ± 0.09
QE CORRECTION COMPUTATIONS			
SLIDE FACE	WIND DIRECTION AND VELOCITY (Set "C" scale index over wind velocity value on "D" scale)	MET CORRECTED QE (Mils) (Set hairline of indicator assembly over met corrected QE value on appropriate "wind elevation"scale)	QE CORRECTION (Mils) (Read on "D" scale under hairline)
Nighttime	Head 3 mph	300 (On "Head Wind Elevation" Scale)	+3.94 ± 0.01
Nighttime	Head 9 mph	650 (On "Head Wind Elevation" Scale)	+35.18 ± 0.07
Nighttime	Tail 4 mph	350 (On "Tail Wind Elevation" Scale)	-6.20 ± 0.01
Nighttime	Tail 8 mph	700 (On "Tail Wind Elevation" Scale)	-32.28 ± 0.06
All other	Head 6 mph	660 (On "Head Wind Elevation" Scale)	+17.93 ± 0.04
All other	Head 20 mph	350 (On "Head Wind Elevation" Scale)	+27.15 ± 0.05
All other	Tail 15 mph	400 (On "Tail Wind Elevation" Scale)	-22.92 ± 0.05
All other	Tail 18 mph	750 (On "Tail Wind Elevation" Scale)	-59.40 ± 0.10

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TABLE XIV

SCALE, GRAPHICAL FIRING; M81

DEFLECTION CORRECTION COMPUTATIONS			
SLIDE FACE	WIND DIRECTION AND VELOCITY (Set "C" scale index over wind velocity value on "D" scale)	MET CORRECTED QE (Mils) (Set hairline of indicator assembly over met corrected QE value on "Lateral Wind Elevation" scale)	DEFLECTION CORRECTION (Mils) (Read on "D" scale under hairline)
Nighttime	Right 2 mph	300	-11.82 ± 0.03
Nighttime	Left 8 mph	650	+70.20 ± 0.13
All other	Right 12 mph	250	-56.44 ± 0.10
All other	Left 8 mph	660	+53.24 ± 0.10
QE CORRECTION COMPUTATIONS			
SLIDE FACE	WIND DIRECTION AND VELOCITY (Set "C" scale index over wind velocity value on "D" scale)	MET CORRECTED QE (Mils) (Set hairline of indicator assembly over met corrected QE value on appropriate "wind elevation" scale)	QE CORRECTION (Mils) (Read on "D" scale under hairline)
Nighttime	Head 3 mph	300 (On "Head Wind Elevation" Scale)	+4.55 ± 0.01
Nighttime	Head 9 mph	650 (On "Head Wind Elevation" Scale)	+40.15 ± 0.08
Nighttime	Tail 4 mph	350 (On "Tail Wind Elevation" Scale)	-7.09 ± 0.02
Nighttime	Tail 8 mph	700 (On "Tail Wind Elevation" Scale)	-36.65 ± 0.07
All other	Head 6 mph	660 (On "Head Wind Elevation" Scale)	+20.43 ± 0.04
All other	Head 20 mph	350 (On "Head Wind Elevation" Scale)	+31.52 ± 0.06
All other	Tail 15 mph	400 (On "Tail Wind Elevation" Scale)	-26.21 ± 0.05
All other	Tail 18 mph	750 (On "Tail Wind Elevation" Scale)	-66.40 ± 0.12

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL-I-45208 and the component inspection requirements of Specification MIL-F-13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot. - Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" specified in Standard MIL-STD-105.

4.2.2 Examination and tests. - Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105, and the inspection level and sampling plans specified in Table XV titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL-S-12937 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

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TABLE XV - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table I with
Sampling Plan Table II-A of Standard MIL-STD-105

		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 1.0 Percent Defective			
101.	Printing and surface quality	3.6	4.5.5
102.	Defective scales	3.7	4.5.6
103.	Graduation error	3.8	4.5.7
104.	Accuracy failure:		
	Type I scale	3.9.1	4.5.8.1
	Type 2 scale	3.9.2	4.5.8.2
MINOR: NONE			

4.2.3 Disposition of non-conforming product. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" specified in Standard MIL-STD-105.

4.3 Control tests. - Three graphical firing scales shall be selected at random from each 50 produced, or from each month's production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
Fabrication	3.2	4.5.1
General specifications	3.3	Visual, MIL-F-13926
Operating temperatures	3.4.1	4.5.2
Storage temperatures	3.4.2	4.5.2, 4.5.4
Humidity	3.4.3	4.5.3, 4.5.4

4.3.1 Control test failure. - Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

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4.4 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIL-I-45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Inspection Equipment", and "Government Furnished Material" requirements of Specification MIL-I-45208.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.2 Equipment used for environmental testing. - All environmental testing equipment used to determine conformance of the environmental requirements for temperature and humidity shall conform to the "Test Facilities" requirements outlined in Specification MIL-F-13926.

4.5 Test methods and procedures.

4.5.1 Fabrication. - The graphical firing scales shall be subjected to a visual, tactile, and dimensional inspection for conformance to the applicable drawings and specifications referenced therein, and to the applicable requirements of this specification. Dimensional inspection shall be performed using standard measuring equipment.

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4.5.2 Storage and operating temperatures. - This test shall be conducted in accordance with Procedure I of Specification MIL-F-13926 except that the temperatures specified in 3.4.1 and 3.4.2 shall apply. While stabilized at each temperature extreme specified in 3.4.1 the firing scale shall be operable when normal thumb pressure is applied to move the indicator of Type I scales, and when applied to the indicator assembly and slide of Type 2 scales. The slide of the Type 2 scale shall not move with normal thumb pressure when pressure is applied (with the thumb and forefinger of the other hand) against the outside edges of the firing scale. Upon completion of the specified exposures and return to standard ambient temperature (60 to 90 degrees Fahrenheit) the firing scale shall be subjected to the tests specified in 4.5.4 and shall conform to the requirements specified therein.

4.5.3 Humidity. - This test shall be conducted in accordance with Procedure I of the humidity test in Specification MIL-F-13926. Upon completion of this test the firing scale shall be subjected to the tests specified in 4.5.4 and shall conform to the requirements specified therein.

4.5.4 Post-environmental. - Subsequent to the tests specified in 4.5.2 and 4.5.3 the firing scale shall be subjected to a visual and tactile examination for defects or damage as specified in 3.5. The firing scale shall then be subjected to the tests specified in 4.5.5 to 4.5.8 inclusive and shall conform to the requirements specified therein.

4.5.5 Printing and surface quality. - This test consists of a visual examination of the face of the firing scale for imperfection of coloring application, letters, numbers, and surface quality, and for stains and other imperfections in accordance with the requirements of 3.6.

4.5.6 Scales. - All firing scales shall be visually inspected for quality of printed graduations and to insure that the colors are in accordance with pertinent drawings in compliance with the requirements of 3.7. The graduations shall then be inspected for accuracy of divisions and sub-divisions by being subjected to the tests specified in 4.5.8 and shall conform to the requirements specified.

4.5.7 Graduation error, "C" and "D" scales - Type 2 scales. - Determine graduation error between the "C" and "D" scales by using a toolmaker's microscope. Lightly clamp the firing scale on the microscope table. Move the center slide until the index of the "C" scale is aligned with the index of the "D" scale and clamp the slide and the rule together lightly to prevent movement. Each graduation on the "C" scale shall be checked for alignment with the opposite graduation on the "D" scale within the tolerance specified in 3.8.1.

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4.5.7.1 Determine the ratio of 2/3 for the "C" scale with respect to the "D" scale using the set-up specified above. Align the index of the "C" scale opposite the 1.5 graduation of the "D" scale and clamp the slide and the rule together lightly to prevent movement. At this setting, a ratio of 2/3 shall be read throughout the firing scale with the opposite graduations in alignment within the tolerance indicated in 3.8.2.

4.5.8 Accuracy tests.

4.5.8.1 Type I scales. - The accuracy tests of the Type I scales are visual tests performed with the aid of a 10X magnifier. Set the hairline of the indicator to the range indicated in the respective tables of 3.9. Read the characteristics for each range as listed in the respective columns of tables I through IX. The firing data characteristics shall be accurate within the tolerances specified in the tables.

4.5.8.2 Type 2 scales. - The accuracy tests for deflection correction and quadrant elevation correction are visually performed with the aid of a 10X magnifier. Align the index of the "C" scale opposite the wind velocity graduation value of the "D" scale indicated in the respective table of 3.9 for the graphical firing scale. Apply pressure with the thumb and forefinger of one hand to the outside edge of the firing scale to hold the scales in this position. Move the hairline of the indicator assembly to the selected quadrant elevation value on the applicable elevation scale and read the deflection correction or quadrant elevation correction on the "D" scale under the hairline of the indicator assembly. The correction readings shall be accurate within the tolerances specified in 3.9.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Specification MIL-S-12937. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.

6.1.1 Type 1 scales. - Graphical firing scales M59, M60, M64, M70, M71, and M85 provide ballistic data pertinent to the application of weapon fire for a particular range and condition.

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6.1.2 Type 2 scales. - Graphical firing scales M74, M78, M79, M80, and M81 are used to determine corrections to deflection and quadrant elevation due to low level winds. Correction readings are determined in accordance with the "Instructions" on the reverse side of the scales.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.

Custodian:

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